VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI -590 014

Flipped class room activity report on

Design of BJT based CE amplifier circuit using TinkerCAD simulation tool

DEPARTMENT

OF

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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S.J.M VIDYAPEETHA ®

S.J.M INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE, New Delhi and Approved by Government of Karnataka)

NAAC Accredited with 'B^{++}' Grade

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1. Introduction

An amplifier is an electronic circuit that increases the strength of a weak signal. Among various amplifier configurations, the Common Emitter **(CE)** amplifier using a Bipolar Junction Transistor **(BJT)** is one of the most widely used due to its high voltage gain and moderate input and output impedance.

In this project, a **BJT CE amplifier** is designed and simulated using **TinkerCAD**, an online circuit simulation tool. The circuit amplifies a small AC input signal into a larger output signal without changing its waveform significantly. Using simulation software helps in understanding circuit behavior, testing components, and avoiding damage to hardware during initial design stages.

2. Objectives

The main objectives of this project are:

- To design and understand the working of a **BJT Common Emitter amplifier**
- To simulate the CE amplifier circuit using **TinkerCAD**
- To study voltage amplification and phase inversion
- To analyze the input and output waveforms
- To gain practical knowledge of transistor biasing and amplification

3. Software Requirements

- **Software:** TinkerCAD Circuits (Online)
- **Web Browser:** Google Chrome / Microsoft Edge / Firefox
- **Internet Connection:** Required
- **Operating System:** Windows / Linux / macOS

4. Steps to Access / Use TinkerCAD

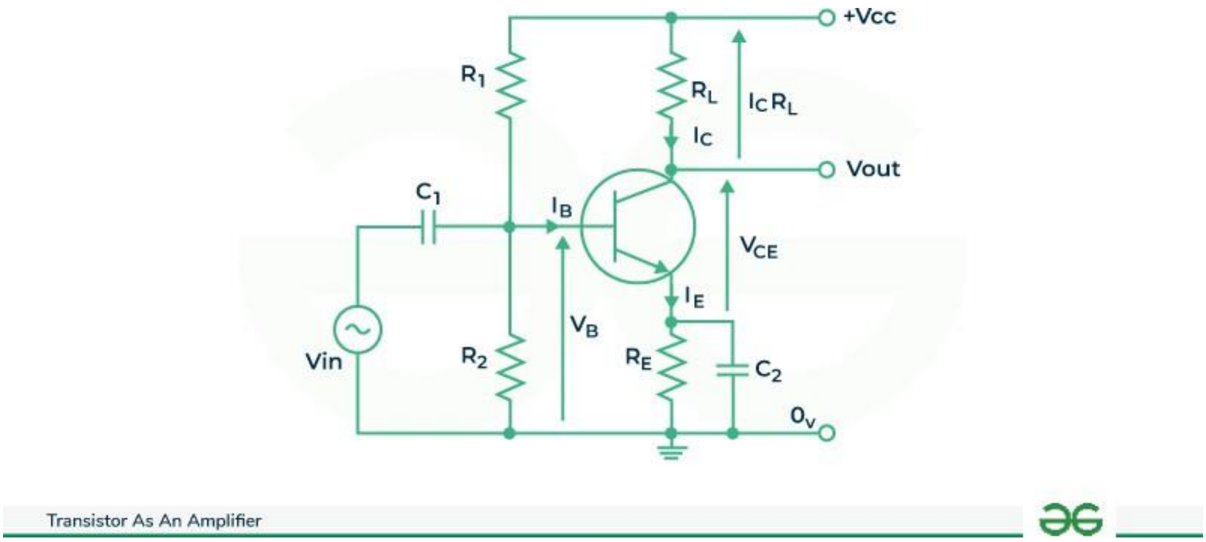
1. Open a web browser.
2. Go to **www.tinkercad.com**
3. Click on **Sign Up** or **Log In**.
4. Choose **Circuits** from the dashboard.
5. Click **Create New Circuit**.
6. Start designing and simulating the circuit.

(TinkerCAD does not require manual installation; it works completely online.)

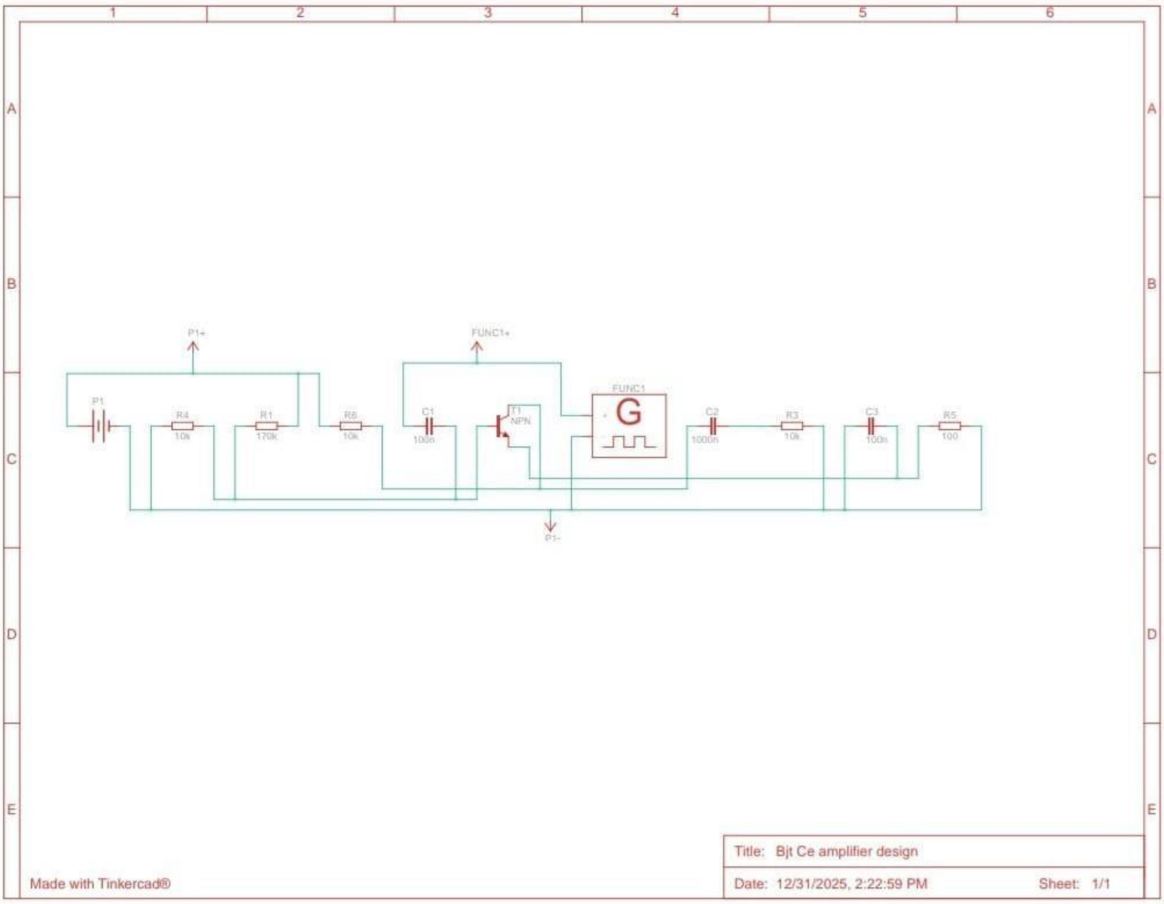
Recommended youtuber – Atara Byte

Link - https://www.youtube.com/@AtaraByte

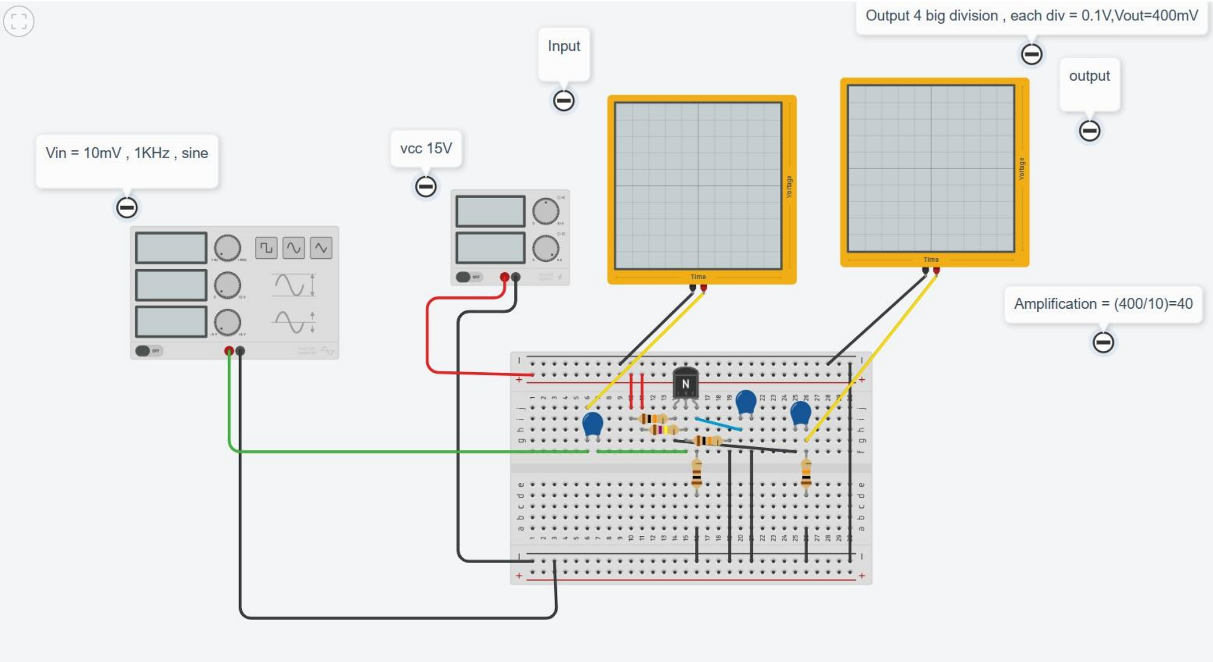
Circuit Diagram



Schematic Diagram



Circuit View



Components Required

Component List

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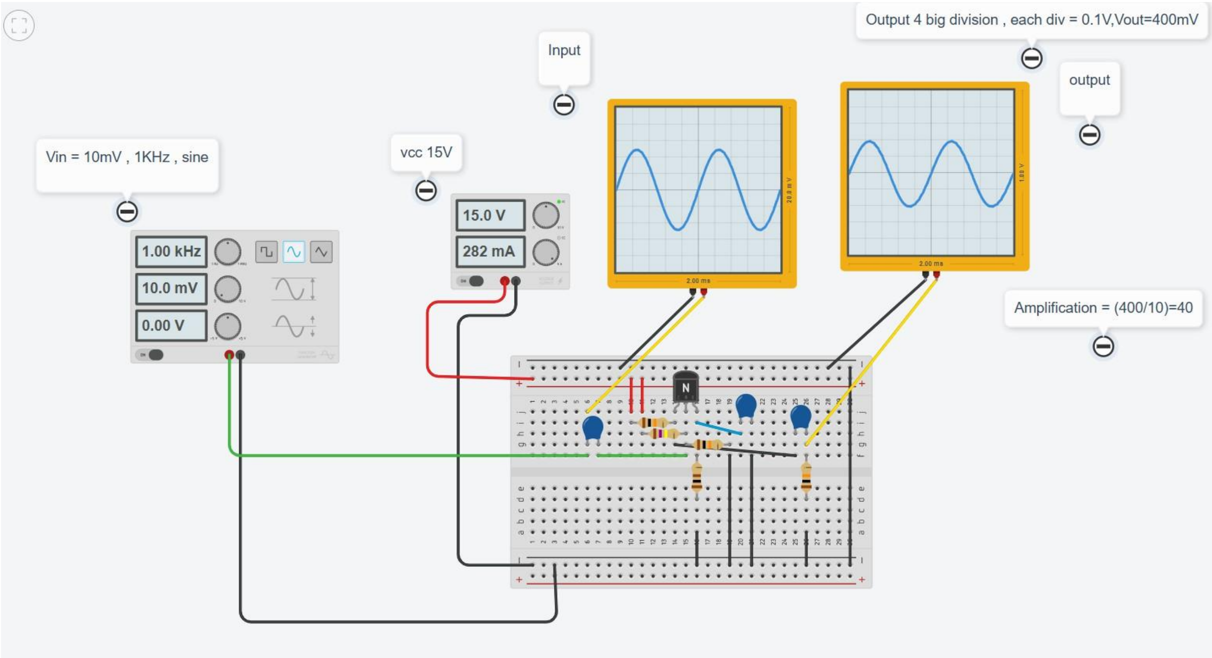
Name	Quantity	Component
T1	1	NPN Transistor (BJT)
R1	1	170 kΩ Resistor
R4 R6 R3	3	10 kΩ Resistor
R5	1	100 Ω Resistor
P1	1	15 , 5 Power Supply
FUNC1	1	1000 Hz, 0.01 V, 0 V, Sine Function Generator
U1 U2	2	0.2 ms Oscilloscope
C1 C3	2	100 nF Capacitor
C2	1	1000 nF Capacitor

Circuit during simulation



5. Observation

- A **10 mV, 1 kHz sine wave** is applied as the input signal.
- The circuit operates with a **15 V DC supply**.
- The BJT is connected in **Common Emitter** configuration.
- The output signal amplitude is **much higher** than the input.
- A **180° phase shift** is observed between input and output.
- Output voltage is approximately **400 mV**
- Av=Vin/Vout
  Av=10mV/400mV
  Av=40
- The calculated **voltage gain is 40**.
- The output waveform is clear and shows proper amplification.
- Coupling capacitors are used at the input and output to **block DC components** and allow only AC signals.
- The waveform shape remains sinusoidal, indicating **minimal distortion** and proper biasing.
- The simulation confirms that the circuit is operating in the **active region** of the transistor.

6. Result and Analysis

The **BJT Common Emitter amplifier** was successfully designed and simulated using TinkerCAD. When a small AC input signal was applied at the base of the transistor, a **larger amplified output signal** was observed at the collector terminal.

The output waveform showed a **180° phase shift** with respect to the input signal, which confirms the characteristics of a CE amplifier. The voltage gain obtained indicates effective amplification. The biasing network maintained stable transistor operation, preventing distortion.

Thus, the simulation results verify the theoretical behavior of a **Common Emitter amplifier**.

7. Conclusion

The **BJT Common Emitter (CE) amplifier** was successfully designed and simulated using **TinkerCAD**. The circuit effectively amplified a small input signal into a larger output signal, demonstrating the principle of transistor-based amplification. The observed **phase inversion of 180°** between the input and output waveforms confirms the characteristic behavior of a CE amplifier.

The simulation helped in understanding the importance of proper biasing, component selection, and circuit configuration for achieving stable amplification. Using TinkerCAD made it easy to design, test, and analyze the circuit without the need for physical components. Overall, this project provided practical knowledge of amplifier operation and reinforced theoretical concepts related to **BJT amplifiers**, making it a valuable learning experience.